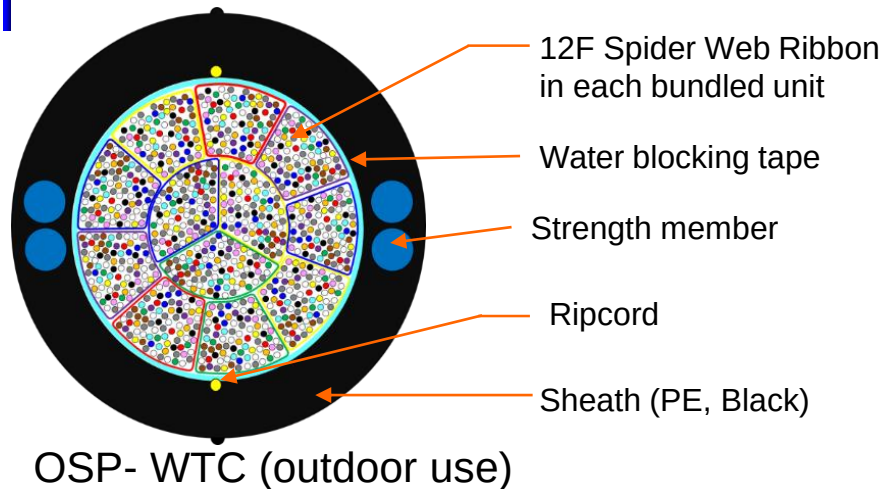


Advantages of SWR/WTC

Typical SWR/WTC (OSP-WTC)



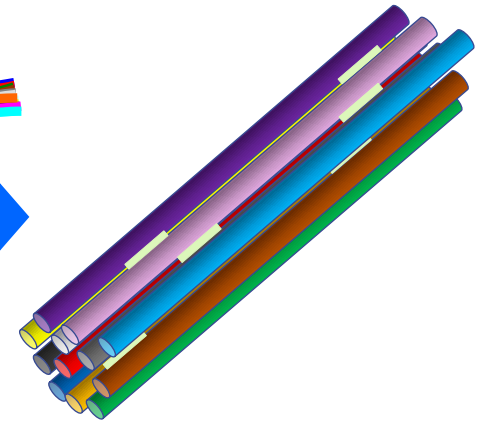
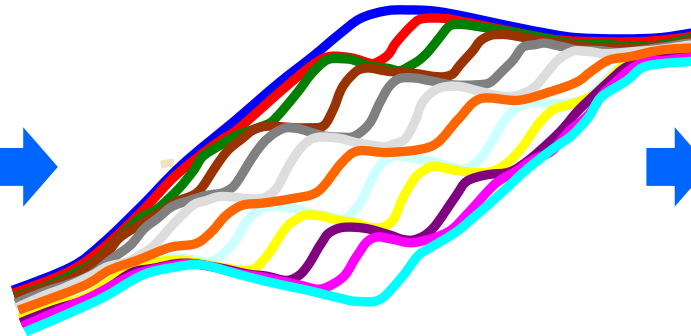
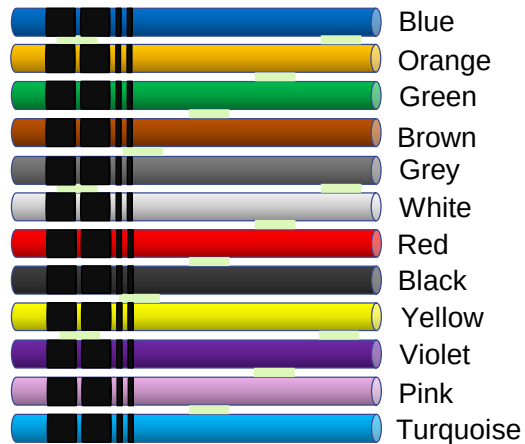
- Full dry (gel-free) construction
- Fully dielectric
- 12 Fiber Spider Web Ribbon (SWR)
- Splicing compatibility with 250µm Ribbons
- UV-resistant, Outdoor duct installation
- ITU-T G.652.D and G.657.A1 compliant

Fiber Count	250µm Fiber type		200µm Fiber type (250µm pitch)	
	Cable dia. (mm)	Cable weight (kg/km)	Cable dia. (mm)	Cable weight (kg/km)
144F	11.0	90	-	-
288F	12.0	105	-	-
432F	13.5	135	-	-
576F	15.0	165	-	-
864F	17.5	215	16.0	185
1728F	23.0	360	21.5	300
3456F	-	-	26.5	450
6912F	-	-	35.0	765

What is SWR and WTC?

SWR : Spider Web Ribbon

- Intermittent bonded ribbon fibers.
- Flexible and Free to form in to bundle structure for high fiber packing density



Fibers are arranged in Parallel
→ Mass Fusion Splicing

Intermittently bonded
→ Flexible and can be separated into 12 single fibers.

Easy to form into bundle
→ High Fiber Packing Density.

What is SWR and WTC?

WTC : Wrapping Tube Cable

- Simple Structure, Easy Access, Light Weight, Smaller Diameter, Faster Deployment
- Comply with International Standards for Mechanical, Environmental Characteristics



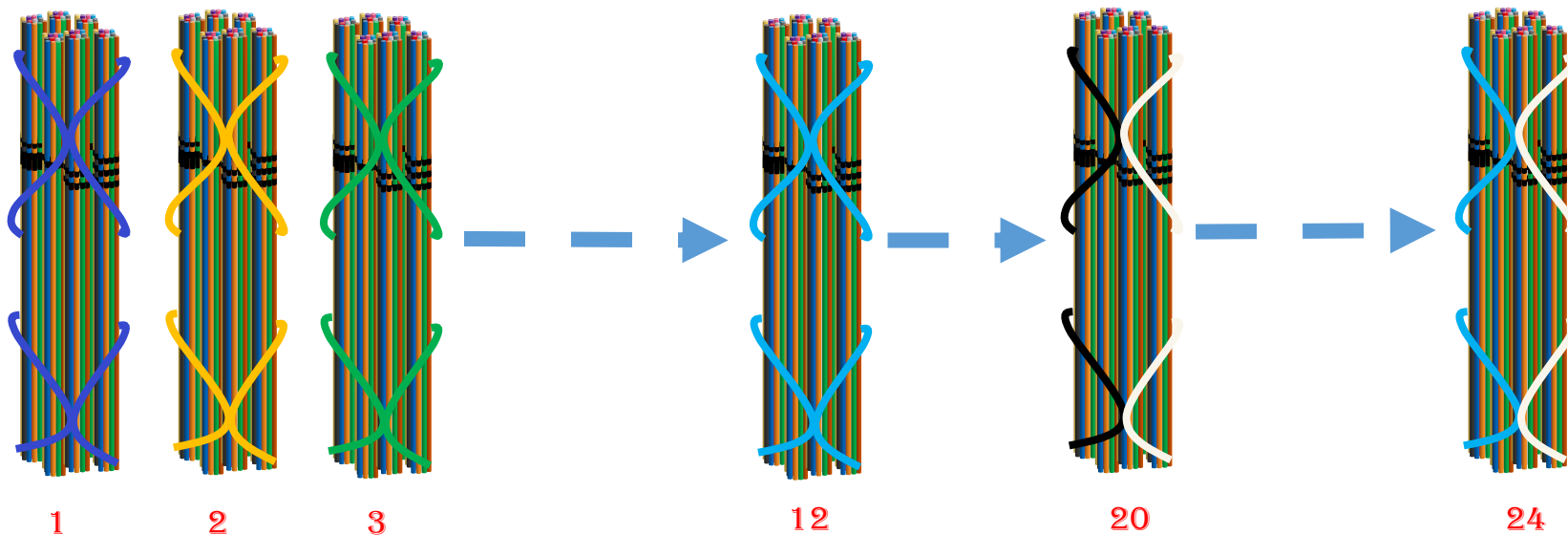
Wrapping tube is a simple technology of wrapping water swellable tape around a group of SWR bundles



What is SWR and WTC?

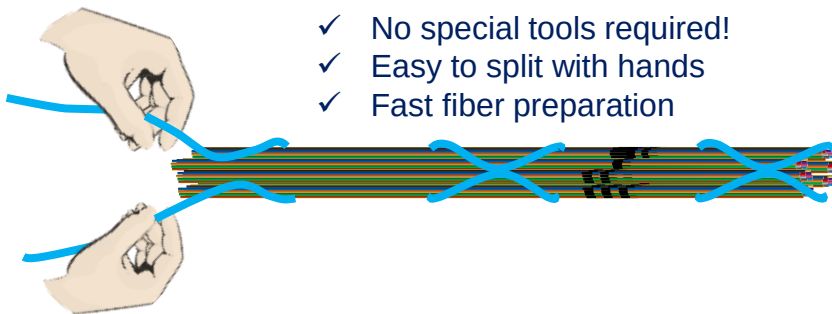
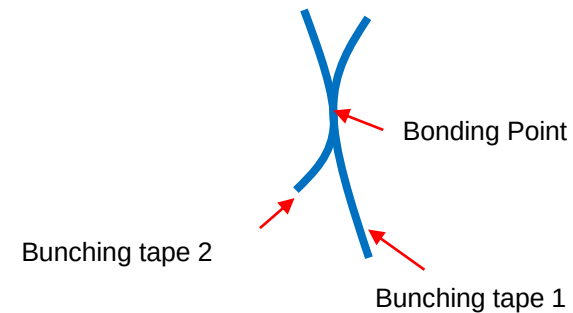
Bundle units

- A certain number of SWRs are bundled if necessary for easy identification and fiber management for higher fiber count cables – typically above 144F cable.
- Bundles are building blocks of high-fiber-count WTC cables.
- Typically there will be 6 numbers of 12F SWRs for cables from 288F to 864F

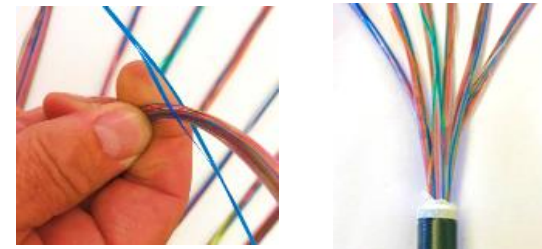


What is SWR and WTC?

- Bundle units are identifiable by color coding of the material with which a group of 12F SWRs are bunched.
- There are 2 (two) bunching tapes that are bonded together at periodic intervals.



- ✓ Retractable, still keeping the structure
- ✓ No gel cleaning



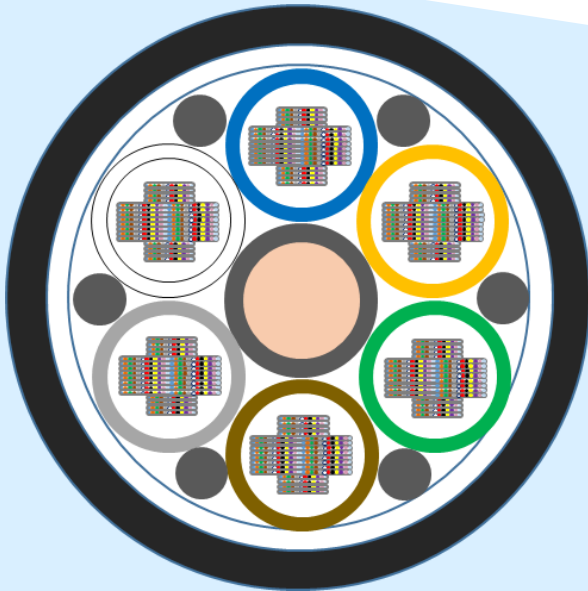
Smaller OD and lighter weight

Manufacture A

Loose tube cable with
rigid ribbon

D = 32mm

W = 750kg/km

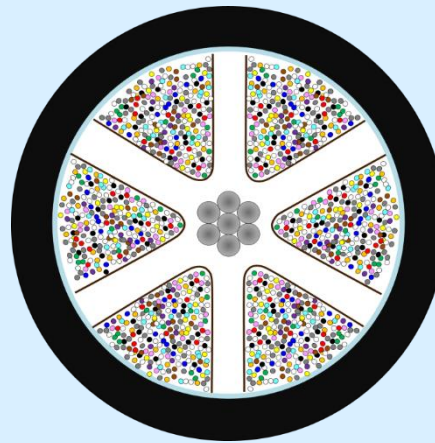


Manufacturer B

Slotted core cable with
partially bonded ribbon

D = 26mm

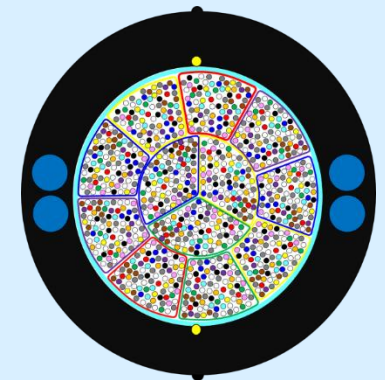
W = 450kg/km



Fujikura
OSP-WTC with SWR

D = **23.0**mm

W = **360**kg/km



*Comparison between 1728 fiber-count cable (250 μ m fiber type)

Smaller OD and lighter weight

Comparison between slotted core cable and OSP-WTC (250 μ m fiber)

Item	Slotted core OSP cable with partially bonded ribbon				OSP-WTC with SWR			
	288F	432F	864F	1728F	288F	432F	864F	1728F
Cable diameter (mm)	17.0	20.0	23.5	29.5	12.0	13.5	17.5	23.0
Cable weight (kg/km)	220	265	365	440	105	135	215	360

OSP-WTCs are **dramatically smaller and lighter** than slotted core cables.

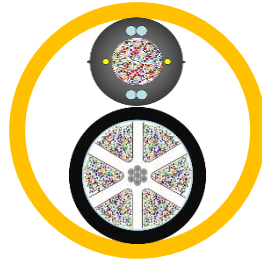
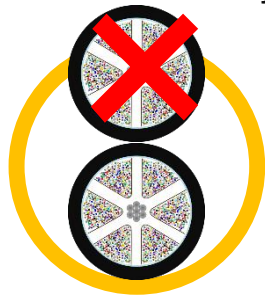
Maximum 30% smaller and **maximum 50% lighter compared** with the same fiber count cable.

What does SWR/WTC bring you?

- **Effective utilization of ducts** for smaller cable outer diameter.

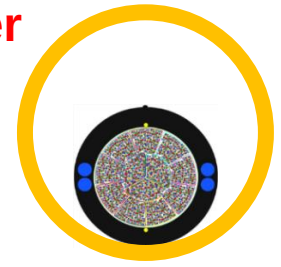
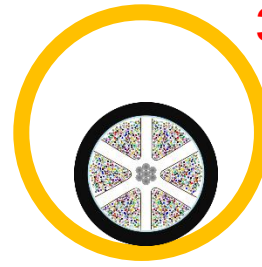
Duct/Hand-Hole Utilization

- Increased fiber capacity within existing infrastructure
- Utilization of duct infrastructure avoids CAPEX of additional duct laying.



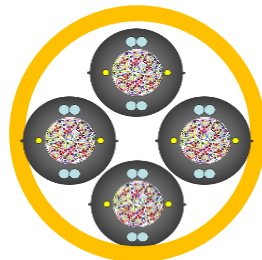
SWR/WTC 288C
Can be added

3 x higher



SWR/WTC 864C
OD : 17.5mm

Slotted core cable with
partially bonded ribbon 288C
OD : 17mm



SWR/WTC
4 x 288C = 1152C

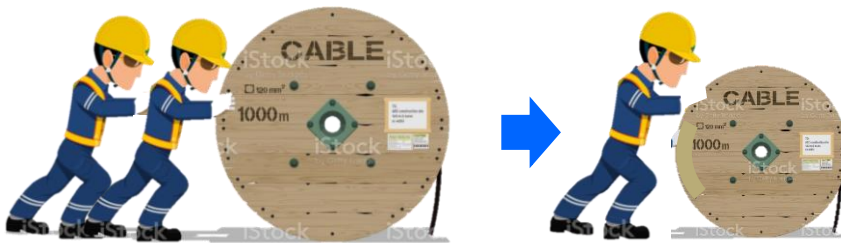
Slotted core cable with
partially bonded ribbon 288C

What does SWR/WTC bring you?

- **Less man power & less transportation cost** for smaller and lighter cable drum.
- **Less connecting points** for longer cable.

Less man power and cost for delivery

- Smaller and lighter cable drum.

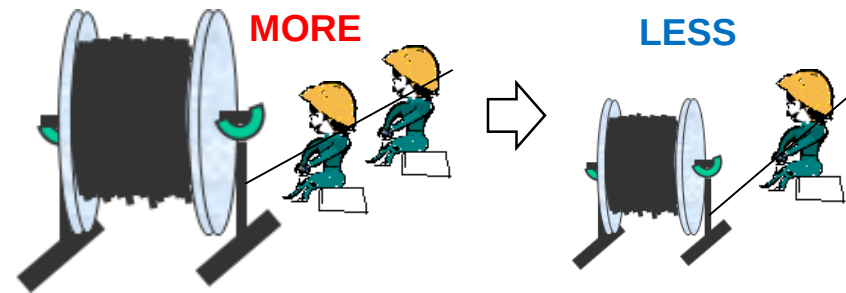


Slotted core cable with partially bonded ribbon 288C

SWR/WTC 288C

Less man power for loading

- Smaller and lighter weight cables offer easier to deploy quickly with longer distance and less craft persons.



Less connection points

- WTC can be wound on a drum as twice longer length due to small cable diameter.
- Longer length per drum reduces the connection points.

What does SWR/WTC bring you?

- **Mass fusion splicing** with excellent compatibility.



**250μm Pitch type of
12F SWR is Splice
Compatible with;**

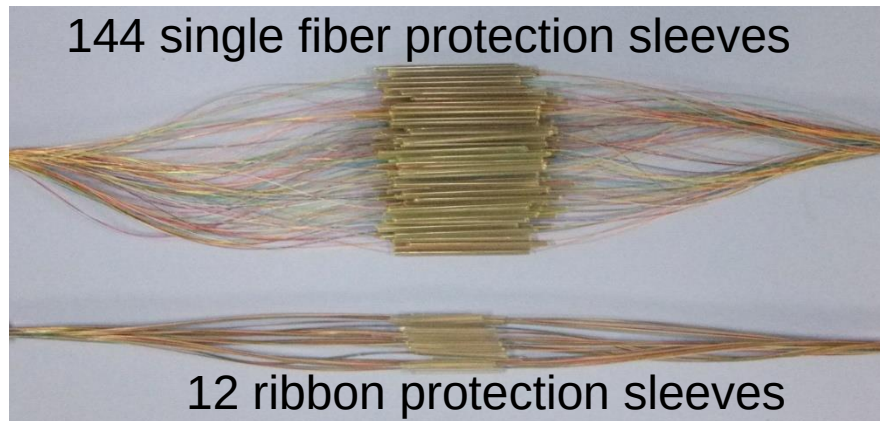
- 250μm Fiber 12F SWR
- 250μm Fiber 8F SWR*
- 250μm Fiber 4F SWR*
- 250μm Fiber 12F Flat Ribbon
- 250μm Fiber 8F Flat Ribbon*
- 250μm Fiber 4F Flat Ribbon*
- Ribbonized Single fibers
- Single fibers

**Forward and Backward
compatibility!**

What does SWR/WTC bring you?

- **Less number of splice protection.**
- **More compact trays / closures** can be introduced.

Less number of splice protection



144 fiber fusion splices / protections
Single fiber LTC vs. SWR.
84% less in volume



Compact trays / closures for
ribbon fibers

What does SWR/WTC bring you?

- Easy installation with various methods.

Duct installation



Direct buried installation



Air-blown installation



Aerial installation



Emerging technology to the world

Developed and Emerging economies Shifting towards Spider Web Ribbon Cables



Supply Chain

Fujikura retains control and ownership of the entire cable supply chain from preform manufacture through to cable assembly.



PREFORM

2 x Manufacturing Locations



FIBER DRAWING

Manufacturing in 2 locations in Japan



SWR

Manufacturing in Japan



WTC

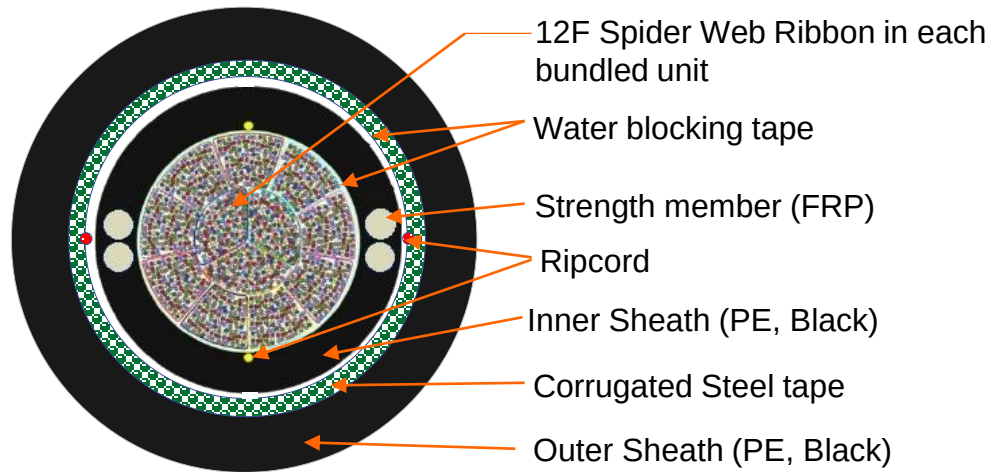
Manufacturing in Japan and the US

Other product lineup

~ With 250 μ m pitch SWR ~

Outdoor Armored SWR/WTC

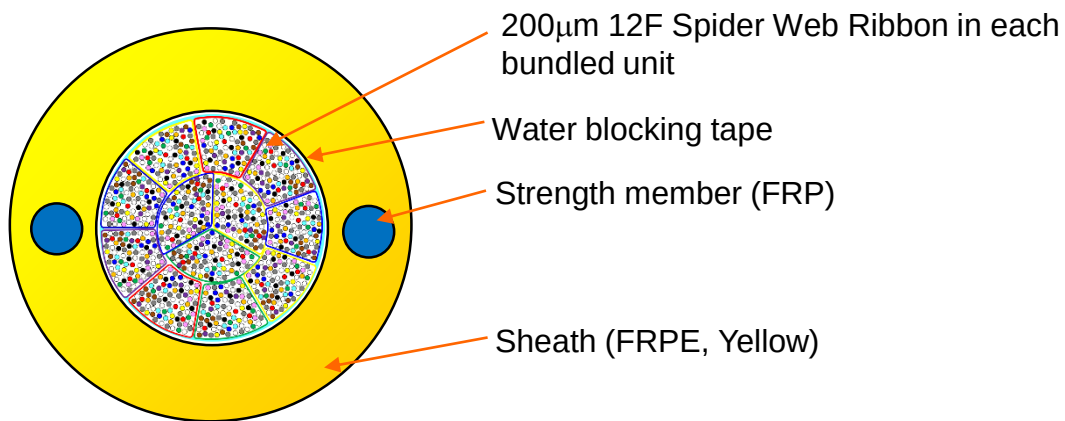
UV Resistant, Rodent Resistant Direct Buried Cable



Fiber Count	250μm Fiber type	
	Cable dia. (mm)	Cable weight (kg/km)
48F	14.5	180
96F	15.0	190
144F	16.0	220
288F	18.0	265
432F	19.5	310
576F	20.5	350
864F	23.0	425
1728F	29.0	625

Indoor SWR/WTC

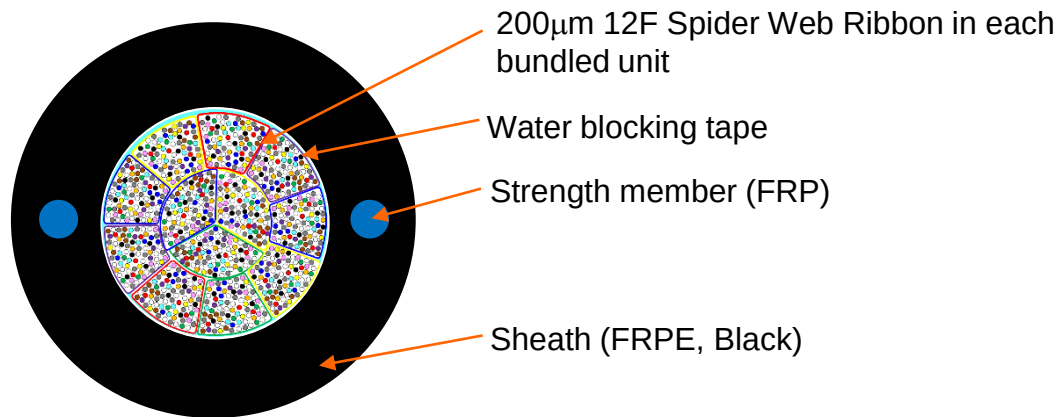
Flame Retardant sheath cables for Indoor wiring



Fiber Count	200µm Fiber type (250µm pitch)	
	Cable dia. (mm)	Cable weight (kg/km)
288F	12.5	155
432F	14.0	195
576F	15.0	220
864F	16.5	265
1152F	17.5	300
1728F	21.5	400
2880F	24.5	540
3456F	27.0	625
6912F	34.5	965

Indoor/Outdoor SWR/WTC

UV Resistant, Flame Retardant cable for Indoor and Outdoor installation



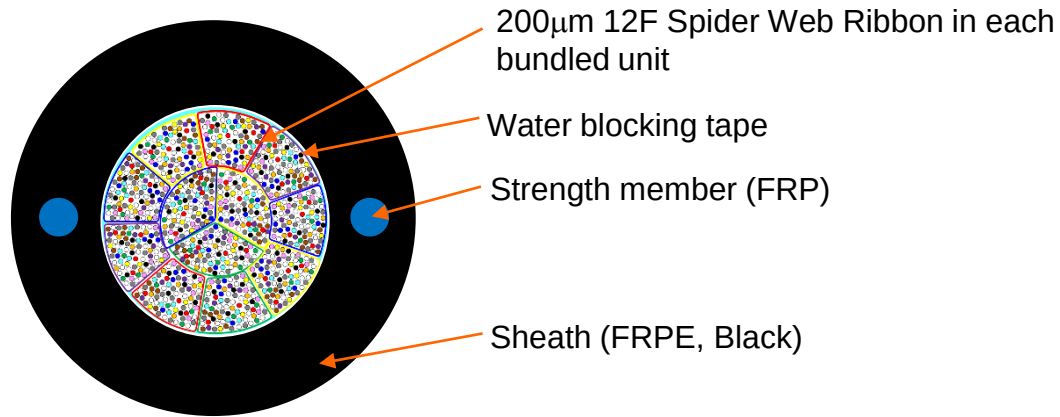
* Yellow color is available as an option, if required



Fiber Count	200µm Fiber type (250µm pitch)	
	Cable dia. (mm)	Cable weight (kg/km)
288F	12.5	155
432F	14.0	195
576F	15.0	220
864F	16.5	265
1152F	17.5	300
1728F	21.5	400
2880F	24.5	540
3456F	27.0	625
6912F	34.5	965

Robust Indoor/Outdoor SWR/WTC

UV Resistant, Robustness for Indoor and Outdoor installation by Jetting.
Tri-rated Flame Retardant for country regulation (Riser, CPR, UKCA)



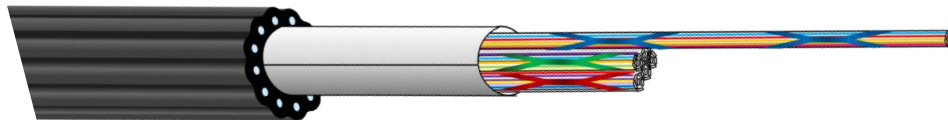
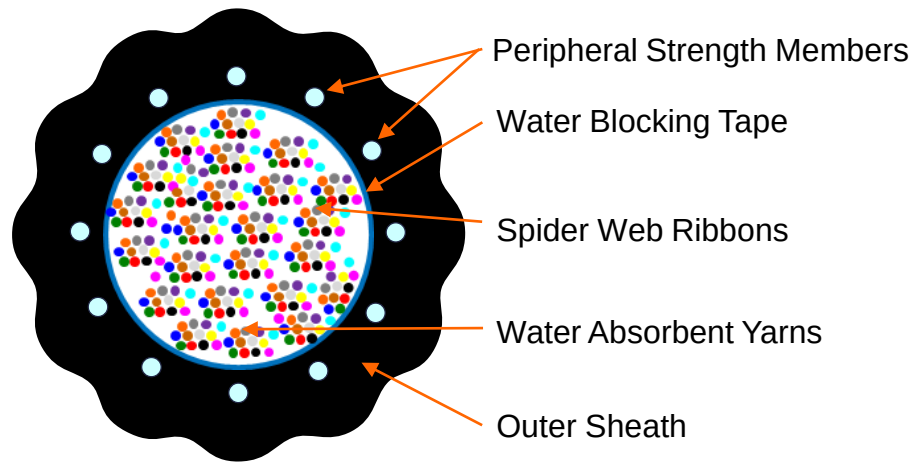
* Yellow color is available as an option, if required



Fiber Count	200μm Fiber type (250μm pitch)	
	Cable dia. (mm)	Cable weight (kg/km)
144F	12.5	165
192F	13.0	175
288F	13.0	190
432F	14.5	215
576F	15.5	240
864F	17.0	290
1152F	18.0	325
1728F	21.5	475
2880F	25.0	615

Air-Blown SWR/WTC

Air Blown cable for Outdoor installation in microducts by Jetting



Fiber Count	200μm Fiber type (250μm pitch)	
	Cable dia. (mm)	Cable weight (kg/km)
48F	6.1	25
72F	6.1	25
96F	6.1	25
144F	6.6	30
192F	7.3	36
288F	8.1	45
432F	9.7	65



Thanks